

Sudhir Gunaseelan

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SUMMARY

Master's graduate in Computer Science focused on Cybersecurity, with hands-on experience in intrusion detection, network traffic analysis, web application security assessment, malware analysis, and vulnerability documentation. Experienced with IDS tools, packet/log analysis, OWASP ZAP, Burp Suite, and ML-based IDS research. Seeking an entry-level Cybersecurity Analyst or SOC Analyst role.

SKILLS

- **Programming:** Python, C, C++, SQL, JavaScript, HTML/CSS
- **Security & Networking:** Wireshark, Kali Linux, Suricata, Snort, OSSEC, tcpdump, Nmap, OWASP ZAP, Burp Suite, Scapy, sslyze, ELK Stack, Metasploit, Network Traffic Analysis, Packet Analysis, Intrusion Detection, Vulnerability Scanning, Log Analysis
- **Malware Analysis & Reverse Engineering:** IDA Freeware, Immunity Debugger, Process Monitor, Process Explorer, INetSim, FakeDNS, x86 Assembly, Static Analysis, Dynamic Analysis
- **Machine Learning & NLP:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Keras, Transformers, LSTM, TF-IDF, SHAP, LIME
- **Web, Systems & Databases:** Flask, Node.js, React, REST APIs, Ubuntu, VirtualBox, Relational Schema Design
- **Tools:** Git, GitHub, Visual Studio Code

EXPERIENCE

Cybersecurity Analyst – Volunteer

April 2026 – Present

Community Dreams Foundation

- Reviewed security controls for CDF-related web applications, including authentication, authorization, input validation, storage access, and public endpoint exposure.
- Used OWASP ZAP, Burp Suite Repeater, manual request review, endpoint inventory, and configuration review to assess public application surfaces and document observations.
- Documented security findings and remediation recommendations related to access control review, upload validation, malformed request handling, and application hardening.

Graduate Grader/TA – COMP 5420/4420 Natural Lang. Proc.

September 2025 – December 2025

University of Massachusetts Lowell – Department of Computer Science

Supervisor: Prof. Hadi Amiri

- Evaluated assignments, midterms, and final projects for a graduate-level NLP course and provided technical feedback to students.
- Developed the course website using jmdoc and ensured consistent and timely grading aligned with instructor guidelines.

Graduate Researcher – KCS Science Masters Program

May 2025 – August 2025

University of Massachusetts Lowell – Kennedy College of Sciences

Supervisor: Prof. Sashank Narain

- Designed a hybrid Intrusion Detection System pipeline combining Bi-LSTM temporal flow modeling, LightGBM tabular classification, and rule-based logic for network traffic classification.
- Preprocessed and analyzed 3 IDS datasets – NSL-KDD, CIC-IDS-2017, and UNSW-NB15 – using imputation, encoding, scaling, and sliding-window generation.
- Tuned confidence thresholds per dataset to classify traffic as Normal, Known Attack, Suspicious, or Zero-Day, routing ambiguous flows for analyst review.
- Achieved >99% precision on high-confidence Known Attack/Normal labels for NSL-KDD and CIC-IDS-2017, and ≈ 89% performance on UNSW-NB15 experiments.
- Applied LIME to LightGBM and SHAP to LSTM outputs to interpret model predictions and map key traffic features into security-relevant categories.

EDUCATION

Master of Science (M.S.) in Computer Science - Cybersecurity

September 2024 – February 2026

University of Massachusetts Lowell, Lowell, MA

GPA: 3.60/4.0

Relevant Coursework: Algorithms, Data Communications, Database, Natural Language Processing, Computer & Network Security, Malware Analysis, Artificial Intelligence, Fundamentals of Robotics, Computer Architecture and Design, Intrusion Detection Systems.

Bachelor of Science (B.S.) in Computer Science

September 2021 – August 2024

University of Massachusetts Lowell, Lowell, MA

GPA: 3.404/4.0 Honors: Cum Laude, Dean's List

PROJECTS

- **Intrusion Detection, SIEM, and Network Security Labs:** Completed 15 IDS exercises configuring Suricata and OSSEC for network and host-based detection, including custom Suricata rules, authentication monitoring, file integrity checks, and alert review. Analyzed PCAPs with tcpdump and Wireshark, including TCP handshakes, SMB/EternalBlue-style traffic, DNS activity, and suspicious scan behavior. Explored alert visualization workflows using ELK Stack/Kibana. **Tech:** Suricata, OSSEC, tcpdump, Wireshark, Nmap, ELK Stack, Linux, DNS (September 2025 – December 2025)
- **Web Application Vulnerability Scanner (Python, Flask):** Coordinated a 3-person team to build a Python/Flask-based vulnerability scanner with 8 modules covering SQL injection, XSS, insecure headers, cookie misconfigurations, SSL/TLS analysis, Heartbleed checks, network scanning, and PDF report generation. Implemented scanning components using Requests, Scapy, sockets, and sslyze, and generated severity-based reports with remediation guidance. **Tech:** Python, Flask, Requests, Scapy, sslyze, sockets, FPDF (February 2025 – May 2025)
- **Malware Analysis and Exploit Development Labs:** Completed 10 malware analysis labs in isolated Windows environments, covering VM setup, static analysis, dynamic analysis, x86 disassembly, IDA Freeware, debugging, Immunity Debugger, manual shellcode generation, Windows malware behavior, and Metasploit payload testing. Reviewed strings, imports, host indicators, process behavior, registry activity, network behavior, buffer overflows, and controlled shellcode execution. **Tech:** IDA Freeware, Immunity Debugger, Process Monitor, Process Explorer, Wireshark, INetSim, FakeDNS, Metasploit, x86 Assembly (February 2025 – May 2025)
- **Wearable Recovery and Stress Monitoring (Python, Transformer, HMM):** Analyzed the LifeSnaps Fitbit dataset by integrating wearable metrics with STAI and PANAS surveys. Built a preprocessing pipeline using imputation, scaling, and sliding windows, then implemented a Transformer model using NumPy alongside a 5-state Gaussian HMM, achieving 88.7% mapped accuracy. **Tech:** Python, NumPy, scikit-learn, Matplotlib (March 2025 – May 2025)
- **Cozmo Robot Programming (Python, OpenCV, Robotics Algorithms):** Implemented Python-based navigation logic for the Cozmo Robot using OpenCV-based colored cube detection, RRT path planning, finite state machines, and Monte Carlo localization with particle filters. Applied software engineering concepts including algorithmic path planning, sensor-based decision-making, and modular task execution for real-time robot behavior. **Tech:** Python, OpenCV, RRT, Monte Carlo Localization, Particle Filters (January 2024 – April 2024)

CERTIFICATIONS

- **Microsoft Cybersecurity Analyst Professional Certificate:** Microsoft / Coursera, In Progress
- **Google Cybersecurity Professional Certificate:** Google / Coursera, In Progress